

Which parts of the previous QAE result survive one-factor changes?

A quantum autoencoder (QAE) robustness study — a follow-up to last meeting's architecture-search result

Baseline finding: physics-informed language-model proposals beat random and greedy circuit search by a wide margin, and a closed feedback loop added a small extra gain.

Four factors tested, one at a time: evaluation budget · qubit count · Hamiltonian · language model.

This is a tested one-factor-slice study, not a complete factorial grid.

Tomoya Hatanaka

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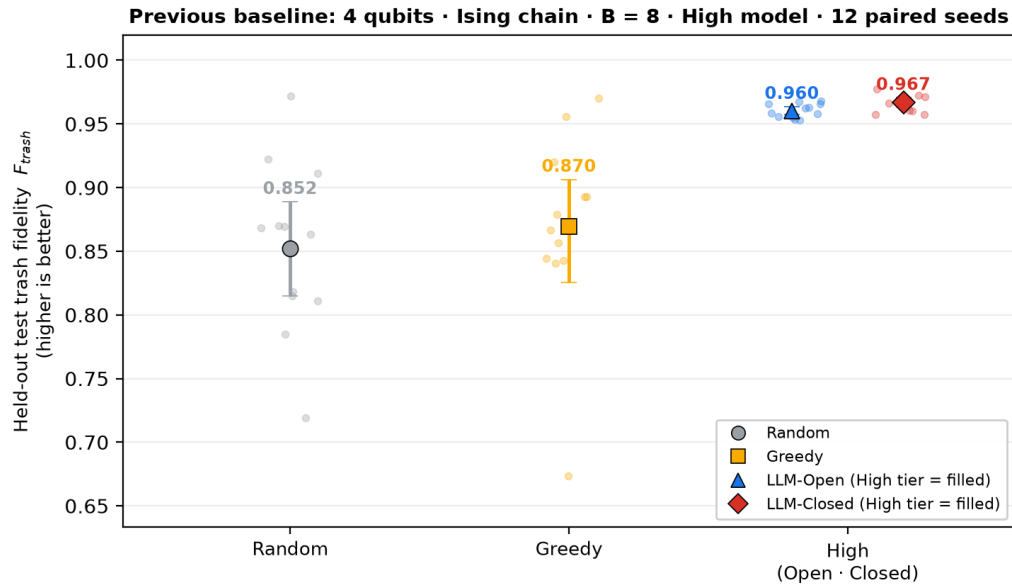
1 · PREVIOUS BASELINE

Where we left off: the previous baseline result

Quantum autoencoder (QAE): compress ground states into fewer qubits — 4 qubits · Ising chain · $B = 8$ · High model tier · 12 paired seeds



Trash fidelity: $F_{\text{trash}} = \langle 00 | \rho_{\text{trash}} | 00 \rangle = P(q_2 q_3 = 00)$, the probability that the trash qubits end in $|00\rangle$. **Higher is better.** Reported on a held-out test set (64 unseen h values never used for training, selection or feedback), read once after selection.



Four search methods; evaluation budget B = the number of candidate circuits trained and evaluated per seed

Random B independent uniformly sampled circuit architectures.

Greedy $B/2$ random starts, then $B/2$ single-structural-change refinements of the best validation candidate.

LLM-Open open-loop: all semantic (physics-informed) proposals are generated BEFORE any validation result is observed.

LLM-Closed closed-loop: semantic initial proposals, then free redesigns of the current best circuit using validation feedback.

Two claims to stress-test

1 Semantic proposals outperform non-semantic search. LLM-Open – Random +0.108 (11/12, $p < 0.001$)

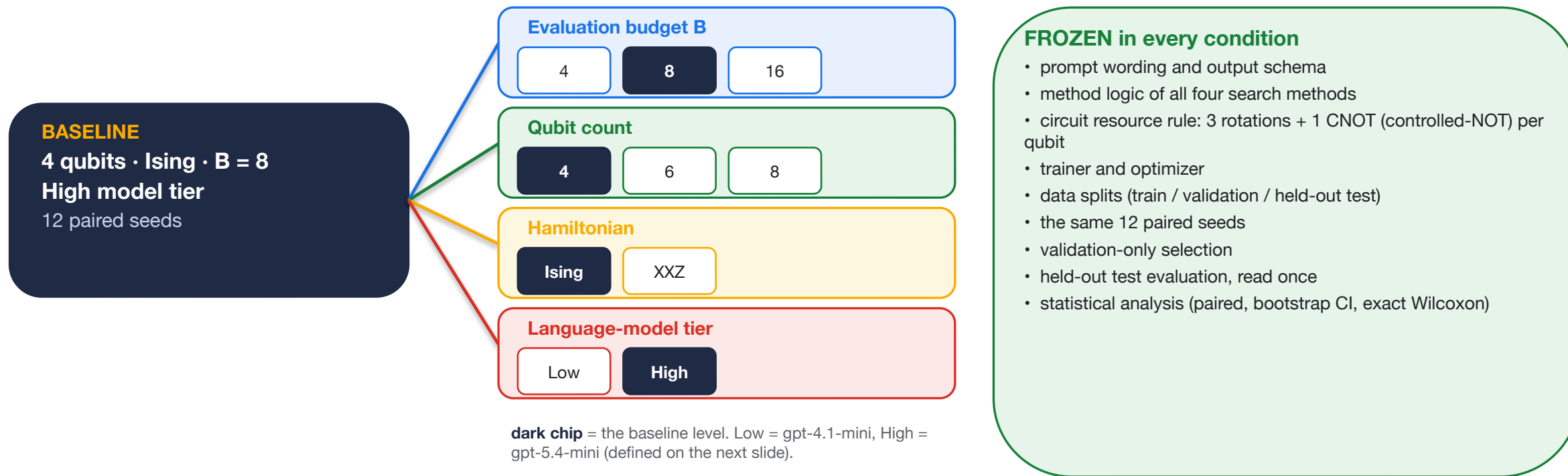
2 Closed-loop redesign gives a smaller additional gain. LLM-Closed – LLM-Open +0.007 (11/12, $p = 0.012$)

Next: do these two claims survive when exactly one condition changes?

Reading: dots = the 12 paired seeds (12 independent draws of training/validation data, shared by all methods); marker = mean; bar = bootstrap 95% confidence interval (CI); differences are per-seed paired, p = exact paired Wilcoxon test. High tier = gpt-5.4-mini.

One factor changes; everything else remains fixed

Four controlled changes leave the baseline; each moves exactly one declared factor and holds the other three at the baseline



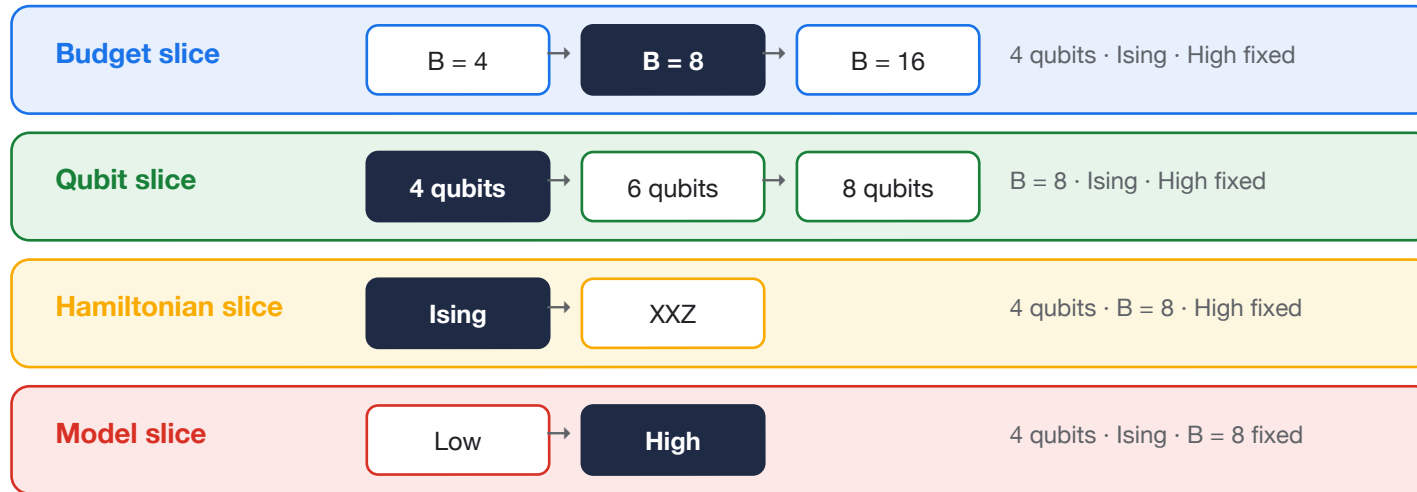
No interaction condition was tested. Each change returns to the baseline before the next factor moves, so nothing is known about, for example, a large budget at 8 qubits, or the Low tier on the XXZ chain. XXZ = the second Hamiltonian family, an XXZ Heisenberg chain $H = \sum (X_i X_{i+1} + Y_i Y_{i+1} + \Delta Z_i Z_{i+1})$, $\Delta \in [0.2, 2.0]$, declared before the run; only its stated physics facts enter the prompt.

Next: because only one factor moves at a time, the results should be read as four tested slices, not as a full grid.

3 · WHAT WAS ACTUALLY TESTED, AND HOW TO READ IT

Four tested slices, one baseline, no empty panels

Every result slide shows one of these four rows; untested combinations are omitted rather than displayed as empty panels



dark chip = the baseline level, present in every slice. Each slice's other conditions were run once, at the fixed settings shown.

Where the Low tier was not run (budget, qubit and Hamiltonian slices), the figures show Random, Greedy and the High Open / Closed results only, and say so; no Low value is invented. The complete Random → Greedy → Low → High grouping appears on the model slide.

Untested combinations are omitted rather than displayed as empty panels.

Model tiers (fixed labels, defined here once)

Low = gpt-4.1-mini **High** = gpt-5.4-mini

The label names the predeclared capability tier, not the score observed in a condition. Open and Closed name the search workflow; Low and High name the model.

How to read every result figure

- small dots = the 12 paired seeds; large marker = mean; vertical interval = bootstrap 95% CI; higher trash fidelity is better
- x-axis groups always in the order Random → Greedy → Low → High
- within a tier: LLM-Open = blue triangle, LLM-Closed = red diamond, side by side
- tier = marker fill: High filled, Low hollow
- thick frame or shaded band = the previous baseline condition
- the small lower panel shows the two key paired differences with 95% CI; filled = $p < 0.05$, hollow = not detected

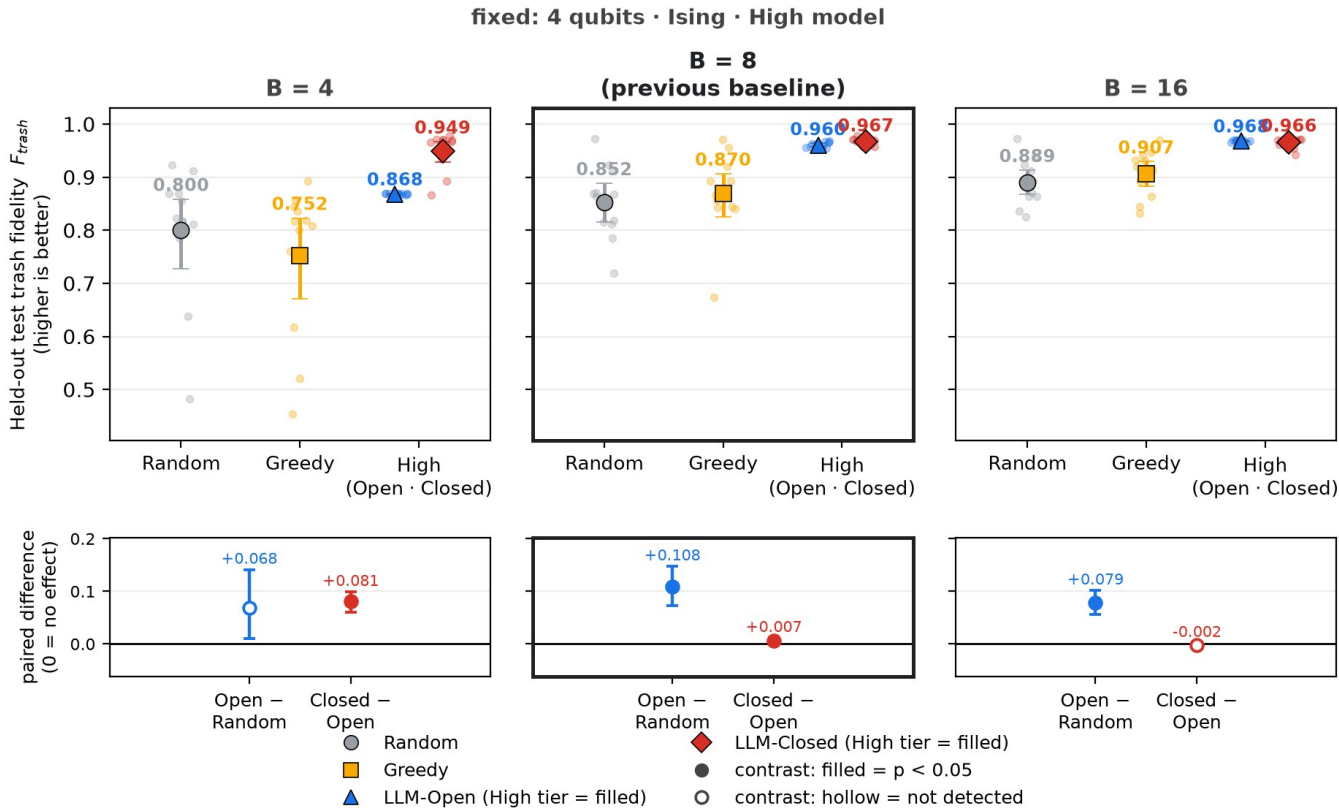
Next: the first slice changes the amount of search: the evaluation budget B.

Random and Greedy do not depend on the language model, so each is one result per condition; Low and High each hold an Open and a Closed result.

4 · BUDGET SLICE

Budget changes the value of feedback

B = 4 → 8 → 16 with 4 qubits, Ising chain and the High model tier fixed; B = 8 (thick frame) is the previous baseline



Semantic proposals stay above non-semantic search at every budget:

LLM-Open – Random +0.068 (8/12, $p = 0.064$) at B = 4, +0.108 (11/12, $p < 0.001$) at B = 8, +0.079 (11/12, $p < 0.001$) at B = 16 — positive throughout, not detected at B = 4 where the budget is too small to separate the arms reliably.

The Closed – Open advantage shrinks as the budget grows: +0.081 at B = 4 → +0.007 at B = 8 → -0.002 at B = 16 (not detected). Free redesign is worth most when the open batch is thinnest; a wide open batch buys the same breadth without feedback.

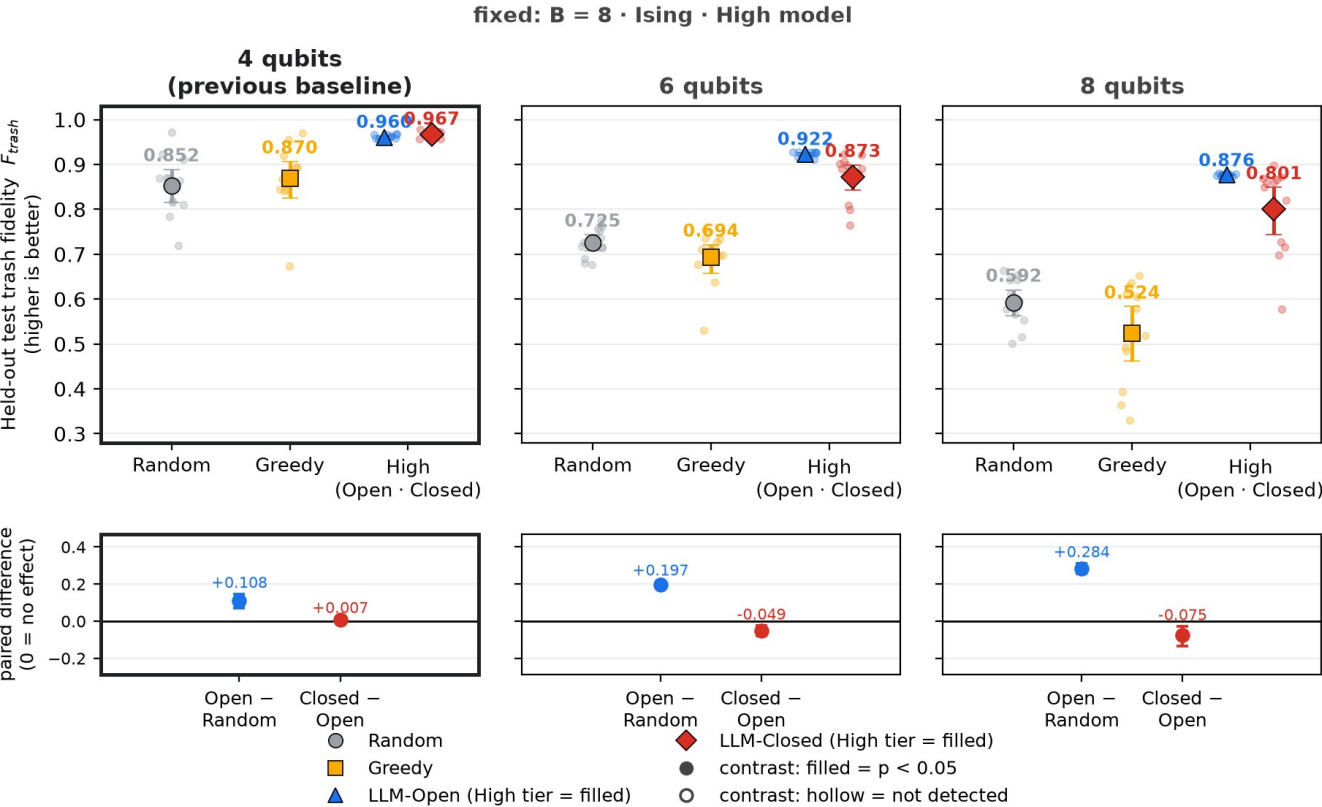
Greedy is not a substitute: Greedy – Random is -0.048 (2/12, $p = 0.016$) at B = 4 — spending half a small budget on single-gate steps costs breadth.

Next: budget changes the amount of search; next we ask whether register width changes the conclusion.

Same trainer, prompts, seeds and selection rule at every budget; adaptive methods always split B as B/2 exploration + B/2 refinement (2+2, 4+4, 8+8).

The semantic advantage grows as the register widens

4 → 6 → 8 qubits with B = 8, Ising chain and the High model tier fixed; 4 qubits (thick frame) is the previous baseline



Finding 1 — absolute fidelity falls as the task widens: Random 0.852 → 0.592; LLM-Open 0.960 → 0.876. Width and circuit size grow together (3 rotations + 1 CNOT per qubit).

Finding 2 — Open stays strong relative to Random; Closed falls below Open: LLM-Open – Random +0.108 → +0.197 → +0.284, significant at all three widths; LLM-Closed – LLM-Open +0.007 → -0.049 → -0.075, reversed at 6 and 8 qubits.

Caveat — proposal validity: the share of closed-loop proposals meeting the gate-count contract (the required 3 rotations + 1 CNOT per qubit) fell from 92% to 70% and 77%; an invalid proposal is spent as a random draw, so part of the reversal is a generation-validity effect.

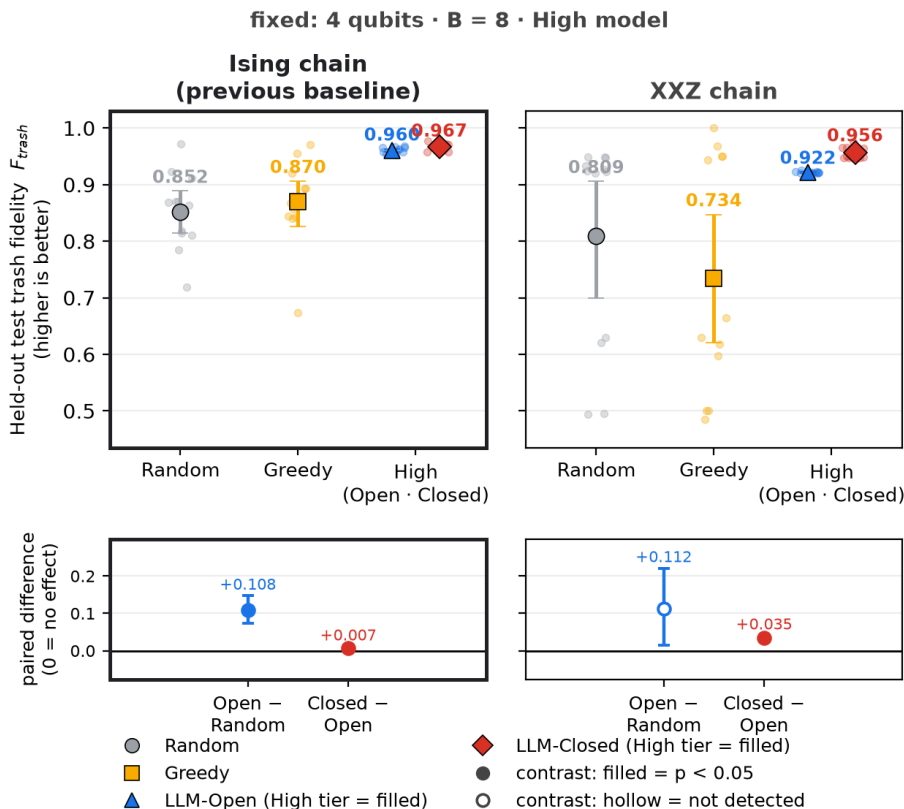
Tested only for the Ising chain and the High tier.

Next: width changes task difficulty; next we change the physical state family while returning to the baseline width.

Same B = 8, prompts (with the qubit count substituted), trainer and seeds at every width; latent and trash registers are always half and half.

The broad ordering survives at the tested XXZ anchor

Ising $\rightarrow$ XXZ with 4 qubits, $B = 8$ and the High model tier fixed; the Ising chain (thick frame) is the previous baseline



The same broad semantic advantage: both semantic methods have higher means than Random on the XXZ chain (LLM-Open 0.922, LLM-Closed 0.956 vs Random 0.809). Per seed, LLM-Closed – Random is +0.147 (11/12, $p < 0.001$); LLM-Open – Random is +0.112 (5/12, $p = 0.791$) — Random's seeds are spread widely here, so the mean gain is not detected as a per-seed effect.

Closed exceeds Open at this anchor: LLM-Closed – LLM-Open +0.035 (12/12, $p < 0.001$), larger than at the Ising baseline.

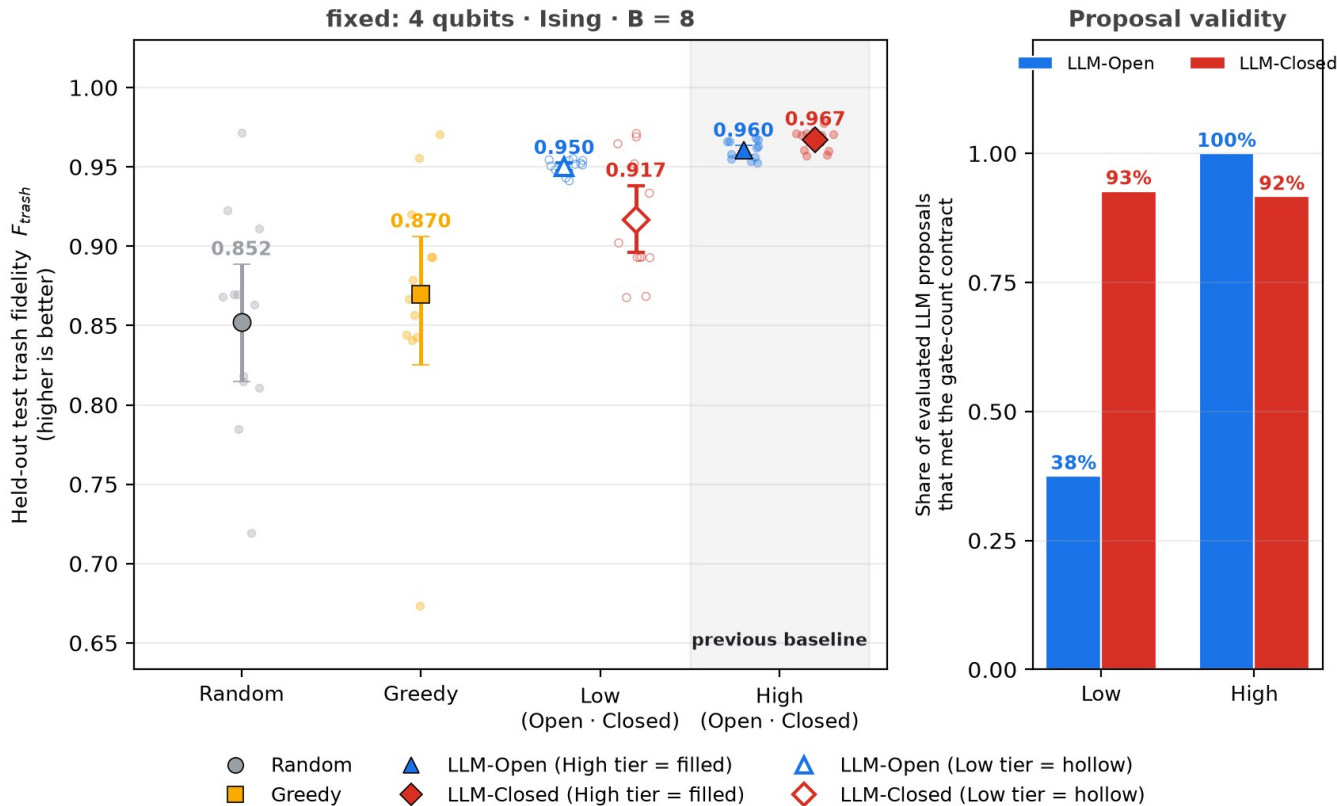
One XXZ point is not the XXZ landscape: it says nothing about how the XXZ result moves with budget, width or model tier — none of those were run for XXZ.

Next: the physics anchor gives one second-family result; the final controlled change is the model itself.

Only the stated Hamiltonian facts change in the prompt (name, formula, parameter range, interaction type); the circuit space, trainer and seeds are identical.

The model tier changes reliability more than the main Open advantage

Low → High with 4 qubits, Ising chain and B = 8 fixed; High (shaded band) is the previous baseline; groups Random → Greedy → Low → High



The Open semantic advantage is similar for both tiers: LLM-Open – Random +0.098 (11/12, $p < 0.001$) for Low and +0.108 (11/12, $p < 0.001$) for High.

The Closed – Open relationship differs between tiers: -0.034 (4/12, $p = 0.052$) for Low versus +0.007 (11/12, $p = 0.012$) for High — the small closed-loop gain seen at the baseline is not present for the Low tier.

Resource-contract validity is a separate failure mode: only 38% of the Low tier's open-batch proposals met the gate-count contract (93% for its closed loop; 100% / 92% for High). Invalid proposals are replaced by random draws, so a validity gap and an architecture-quality gap are entangled in the plotted score.

Low and High are fixed tier labels; the plotted score is not what defines them.

Next: the four slices can now be combined to separate robust effects from fragile ones.

Identical prompt bytes, temperature, retry policy and token limit for both tiers; only the model name changes. Random and Greedy are the same runs in both groups.

What survived across the tested slices?

One row per tested slice; wording is evidence-calibrated — robust, condition-dependent, not detected, reversed, not tested

Slice	Open versus Random	Closed versus Open	Proposal validity
Budget (B = 4, 8, 16)	condition-dependent: not detected (B = 4) · robust (B = 8) · robust (B = 16)	condition-dependent: robust (B = 4) · robust (B = 8) · not detected (B = 16)	stable (≥ 92%)
Qubit count (4, 6, 8)	robust	condition-dependent: robust (4) · reversed (6) · reversed (8)	degraded: Closed at 6 qubits 70%
Hamiltonian (Ising, XXZ)	condition-dependent: robust (Ising) · not detected (XXZ)	robust	stable (≥ 92%)
Model tier (Low, High)	robust	condition-dependent: not detected (Low) · robust (High)	degraded: Low Open 38%

Not tested: width and Hamiltonian changes for the Low tier; any change away from the baseline for XXZ; every two-factor combination.

Across the tested one-factor slices

- the semantic Open advantage is the most robust finding — same sign in every slice, significant in most conditions;
- the additional closed-loop advantage is budget-, width-, Hamiltonian- and model-dependent — it is not a general result;
- proposal validity can confound observed method differences: an invalid proposal is spent as a random draw.

Key limitations

- no interactions; only one non-Ising anchor
- Low tested only at the baseline physical condition
- 12 paired seeds: a "not detected" cell is not evidence of no effect
- noiseless state-vector simulation
- qubit count and circuit size scale together

Next: ranking methods at a fixed budget answers the wrong question for practice — how much budget is actually required?

"robust" = same sign as the baseline and $p < 0.05$; "reversed" = $p < 0.05$ with the opposite sign; "not detected" = $p \geq 0.05$; "condition-dependent" = the verdict differs across the slice's levels.

Next: how much budget is actually required?

Three statements from the tested slices, then a targeted boundary-finding experiment rather than a factorial sweep

Robust

Physics-informed Open proposals remain better than non-semantic search across the tested one-factor slices.

Fragile

The closed-loop advantage is not a general result: it changes with budget, width, Hamiltonian and model tier.

Operational bottleneck

Valid circuit generation must be separated from architecture quality — an invalid proposal is spent as a random draw.

Why this matters: every comparison so far ranks methods at a fixed B. The question that decides whether any of this is worth using is the reverse — the budget each method needs to reach the accuracy the application requires.

Next experiment — the minimum budget for a target fidelity

Define before running: a target validation fidelity F_{target} (candidate targets 0.95 and 0.99). B_{min} = the smallest budget that reaches F_{target} in at least 10 of the 12 paired seeds. B_{min} is determined on validation data; the held-out test is read only after B_{min} is selected.

API-minimising procedure

- 1 Reanalyse the existing logs first — no new calls.
- 2 Build best-so-far validation-fidelity curves versus candidate-evaluation count from those logs.
- 3 Estimate whether each target is already reached by $B = 4, 8$ or 16 .
- 4 Identify only the conditions whose B_{min} remains unresolved.
- 5 Run new LLM calls only at the nearest bracketing budget for those unresolved conditions.
- 6 Preserve the same 12 paired seeds for any new confirmatory run.
- 7 Report an interval for B_{min} when the exact minimum cannot be identified without excessive API cost.

Not proposed: a blind full sweep over every Hamiltonian, qubit count, model, method and budget. The practical next step is targeted boundary-finding, not factorial completion.